

DAY 08- NETWORK CHALLENGE

NAME:- RISHABH RAJ

TASK1:- NETWORK IDENTITY

- IPv4 Address:- 10.83.207.182
- Subnet Mask:- 255.255.255.255
- Default Gateway:- fe80::415:5dff:fe49:3r44%8
- DNS server:-10.83.217.244
- MAC/Physical Address:-
- DHCP:- Enabled

TASK5:- NETWORK PATH INVESTIGATION

- Run this command:- traceroute google.com
- Total hop:- 14
- Number of visible hop:- 09
- First visible hop:- 1

What the first hop commonly represents:- It commonly represents the local router/default gateway used to send traffic outside the local network.

Purpose of traceroute:- It shows the sequence of network hop/routers toward a destination and helps diagnose routing or connectivity problems.

Screenshots:

```
C:\Users\satya>tracert google.com

Tracing route to google.com [2404:6800:4002:81d::200e]
over a maximum of 30 hops:

  1    5 ms    3 ms    2 ms  2409:40d4:117:eddb::cf
  2    *      *      *      Request timed out.
  3   77 ms   208 ms  337 ms 2405:200:5216:0:3925::1
  4  104 ms   111 ms   96 ms 2405:200:801:1600::780
  5    *      *      *      Request timed out.
  6   13 ms   18 ms   15 ms 2405:200:812:1513:61::7
  7    *      *      *      Request timed out.
  8    *      *      *      Request timed out.
  9    *      *      *      Request timed out.
 10    *      *      *      Request timed out.
 11   24 ms   18 ms   16 ms 2001:4860:1:1::17ae
 12   31 ms   27 ms  163 ms 2001:4860:0:1::7819
 13   70 ms   40 ms   40 ms 2001:4860:0:1::5e49
 14   22 ms   36 ms   23 ms del11s18-in-x0e.1e100.net [2404:6800:4002:81d::200e]

Trace complete.
```

TASK4:- DNS INVESTIGATION

Run this command:- nslookup google.com

- IP address:- 142.250.29.102

Purpose of DNS:-

DNS translate domain name such as google.com into IP addresses so computers can locate and communicate with the destination.

Screenshots:

```
rishabhraj@Rishabhs-MacBook-Air ~ % nslookup google.com
Server:                2409:40d4:117:eddb::cf
Address:               2409:40d4:117:eddb::cf#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.29.102
Name:   google.com
Address: 142.250.29.101
Name:   google.com
Address: 142.250.29.139
Name:   google.com
Address: 142.250.29.100
Name:   google.com
Address: 142.250.29.113
Name:   google.com
Address: 142.250.29.138

rishabhraj@Rishabhs-MacBook-Air ~ %
```

File display

TASK2 & 3:- QUIZ

- Completed Quiz.

TASK6:- FLAG

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